in a given interval of time than would a beam of light, and hence the path it follows through spacetime must make an angle with the centerline of the diagram (the line running through the center of the loaf from crust to crust) that is less than 45 degrees. Moreover, Einstein showed that the time slices for an observer moving with velocity *v*—all of space at one moment of such an observer's time—have an equation (assuming one space dimension for simplicity) given by $t_{moving} = \gamma(t_{stationary} - (v/c^2)\, x_{stationary})$, where $\gamma = (1 - v^2/c^2)^{-1/2}$, and c is the velocity of light. In units where c = 1, we note that $v < 1$ and hence a time slice for the moving observer—the locus where t_{moving} takes on a fixed value—is of the form $(t_{stationary} - vx_{stationary}) =$ constant. Such time slices are angled with respect to the stationary time slices (the loci of the form $t_{stationary}$ = constant), and because $v < 1$, the angle between them is less than 45 degrees.

10. For the mathematically inclined reader, the statement being made is that the geodesics of Minkowski's spacetime—the paths of extremal spacetime length between two given points—are geometrical entities that do not depend on any particular choice of coordinates or frame of reference. They are intrinsic, absolute, geometric spacetime features. Explicitly, using the standard Minkowski metric, the (timelike) geodesics are straight lines (whose angle with respect to the time axis is less than 45 degrees, since the speed involved is less than that of light).

11. There is something else of importance that all observers, regardless of their motion, also agree upon. It's implicit in what we've described, but it's worth stating directly. If one event is the cause of another (I shoot a pebble, causing a window to break), all observers agree that the cause happened *before* the effect (all observers agree that I shot the pebble *before* the window broke). For the mathematically inclined reader, it is actually not difficult to see this using our schematic depiction of spacetime. If event A is the cause of event B, then a line drawn from A to B intersects each of the time slices (time slices of an observer at rest with respect to A) at an angle that is *greater* than 45 degrees (the angle between the space axes—axes that lie on any given time slice—and the line between A and B is greater than 45 degrees). For instance, if A and B take place at the same location in space (the rubber